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Course : MATH6030001 - Linear Algebra

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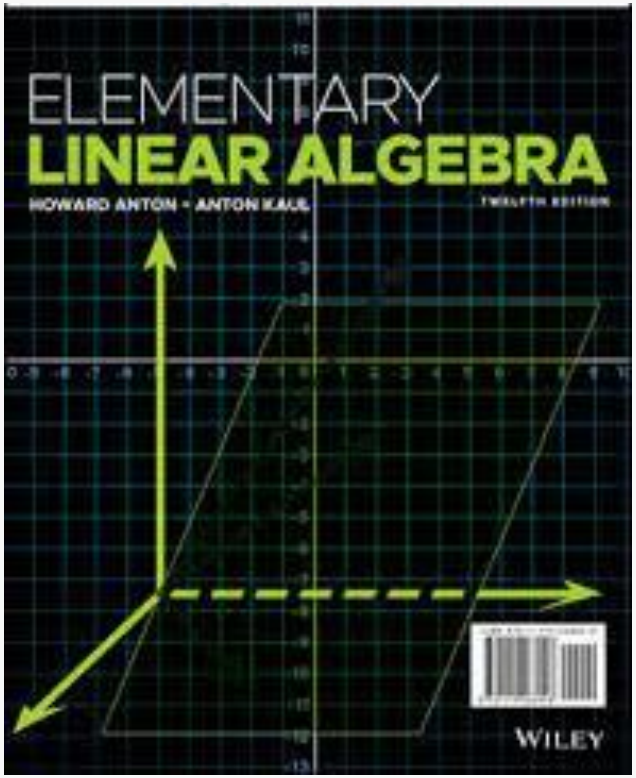
Systems of Linear Equations

Session 01-02

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Acknowledgement



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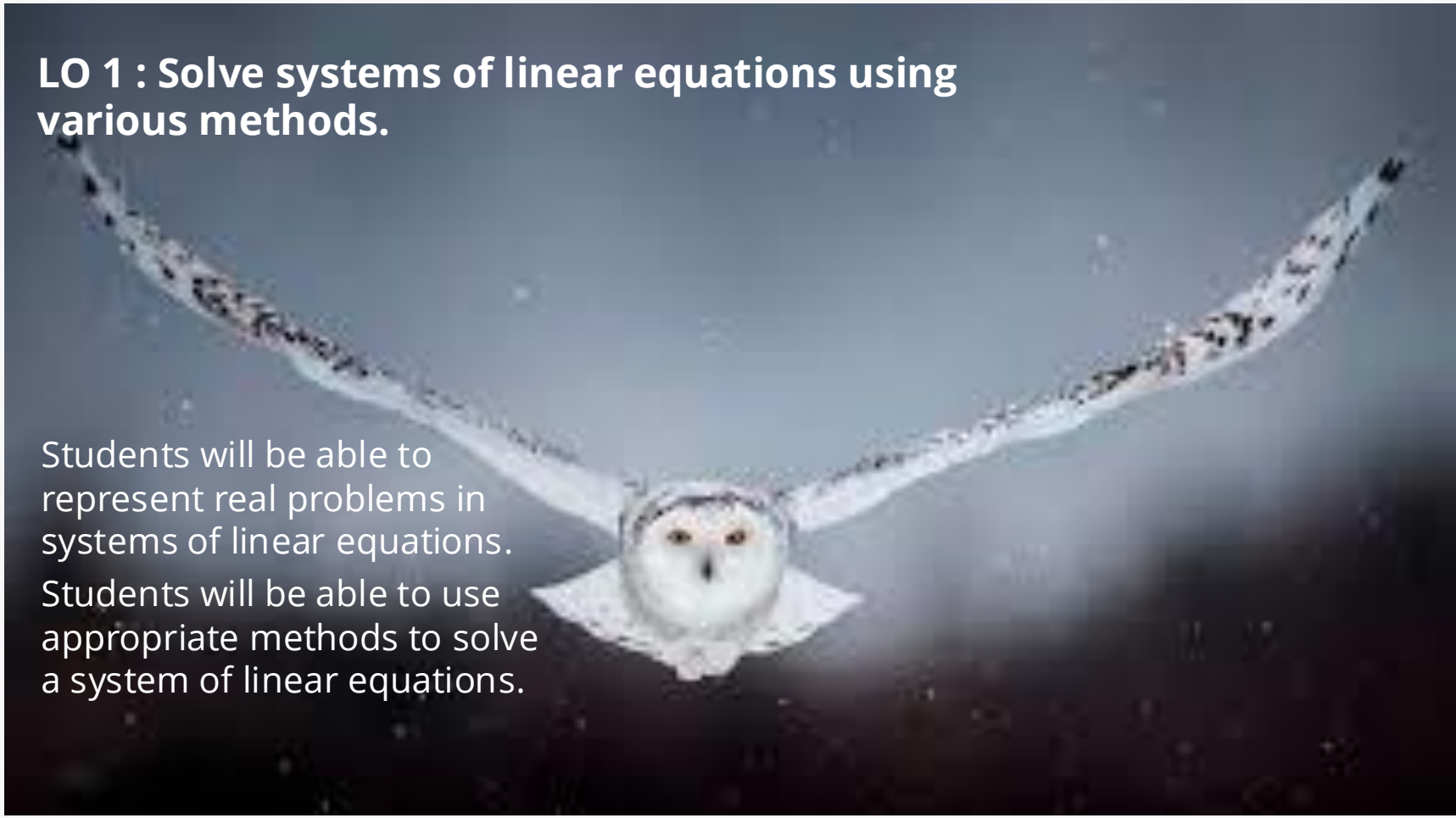
Howard Anton and Anton Kaul. *Elementary linear algebra: Applications version (12th Edition)*. John Wiley & Sons, 2019.

Section 1.1-1.2

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Learning Objectives



LO 1 : Solve systems of linear equations using various methods.

Students will be able to represent real problems in systems of linear equations.

Students will be able to use appropriate methods to solve a system of linear equations.

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Contents



1 System of linear equations

2 Solution

3 Gaussian and Gauss-Jordan Elimination

4 Homogeneous System of Linear Equations

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Definition

Linear Equations in several variables are equations in the form of polynomials which have degrees one or zero and there is no multiplication between variables.

A **linear equation** in the variables $x_1 \ldots x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

Where $a_1, \ldots, a_n$, and b are constant.

Example :

$2x_1 - x_2 + 1.5x_3 = 8 \ldots (1)$

$2x_1\,x_2 + x_2 = 2 \quad \ldots \quad (2)$

$2x_1 - x_2 + \sqrt{x_3} = 8 \quad \ldots \quad (3)$

The equation (2) and (3) are not linear because of the presence of $x_1\,x_2$ in the first equation and $\sqrt{x_3}$ in the second

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Definition

A **system of linear equations** (or a linear system) is a collection of one or more linear equations involving the same variables —say, $x_1 \ldots xn$.

Example:

$$2x_1 - x_2 + 1.5x_3 = 8$$
$$x_1 - 4x_3 = -7$$

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SOLUTION OF LINEAR SYSTEM

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A solution of a linear system in n unknowns $x_1, x_2, \ldots, x_n$ is is a sequence of n numbers

$s_1, s_2, \ldots, sn$ for which the substitution

$x_1 = s_1, x_2 = s_2, \ldots x_n = s_n$

Example : Determine solution for system of linear equations as follows :

$$5x + y = 3$$
$$2x - y = 4$$

Solution : the system has the solution $x = 1, y = -2$

A system of linear equations has

1. no solution, or
2. exactly one solution, or
3. infinitely many solutions.

A system of linear equations is said to be consistent if it has either one solution or infinitely many solutions; a system is inconsistent if it has no solution.

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Linear Systems with Two Unknown

Linear systems in two unknowns arise in connection with intersections of lines. For **example**, consider the linear system

$$a_1x + b_1y = c_1$$
$$a_2x + b_2y = c_2$$

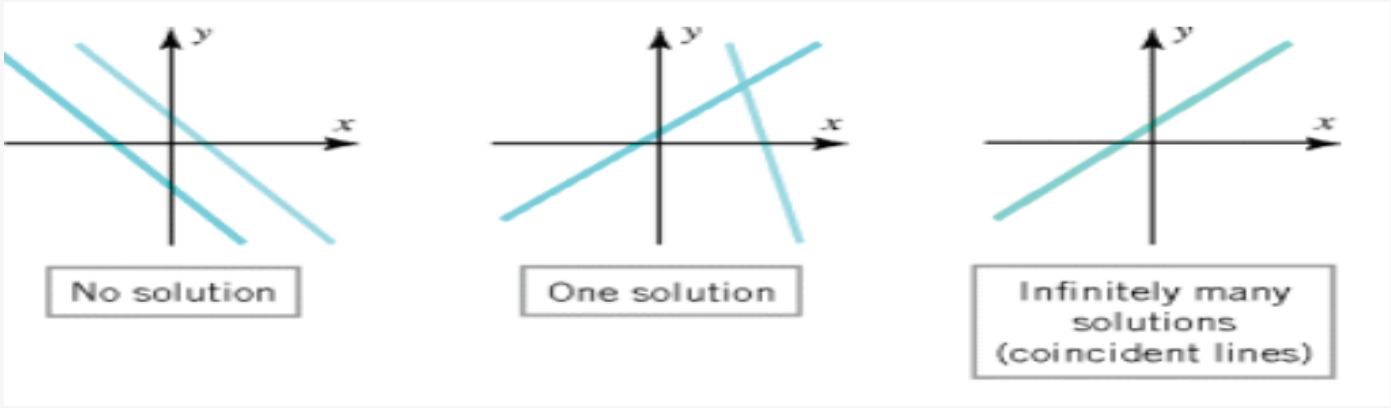
Each solution (x,y) of this system corresponds to a point of intersection of the lines, so there are three possibilities :

1. no solution, if the lines may be parallel and distinct,
2. exactly one solution, if the lines may intersect at only one point.
3. infinitely many solutions, if the lines may coincide, in which case there are infinitely many points of intersection.

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Linear Systems with Two Unknown



Examples :

A linear system with no solution

$$x + y = 4$$
$$3x + 3y = 6$$

A linear system with one solution

$$5x + y = 3$$
$$2x - y = 4$$

A linear System with infinitely many solutions

$$4x - 2y = 1$$
$$16x - 8y = 4$$

Linear Systems with Three Unknowns

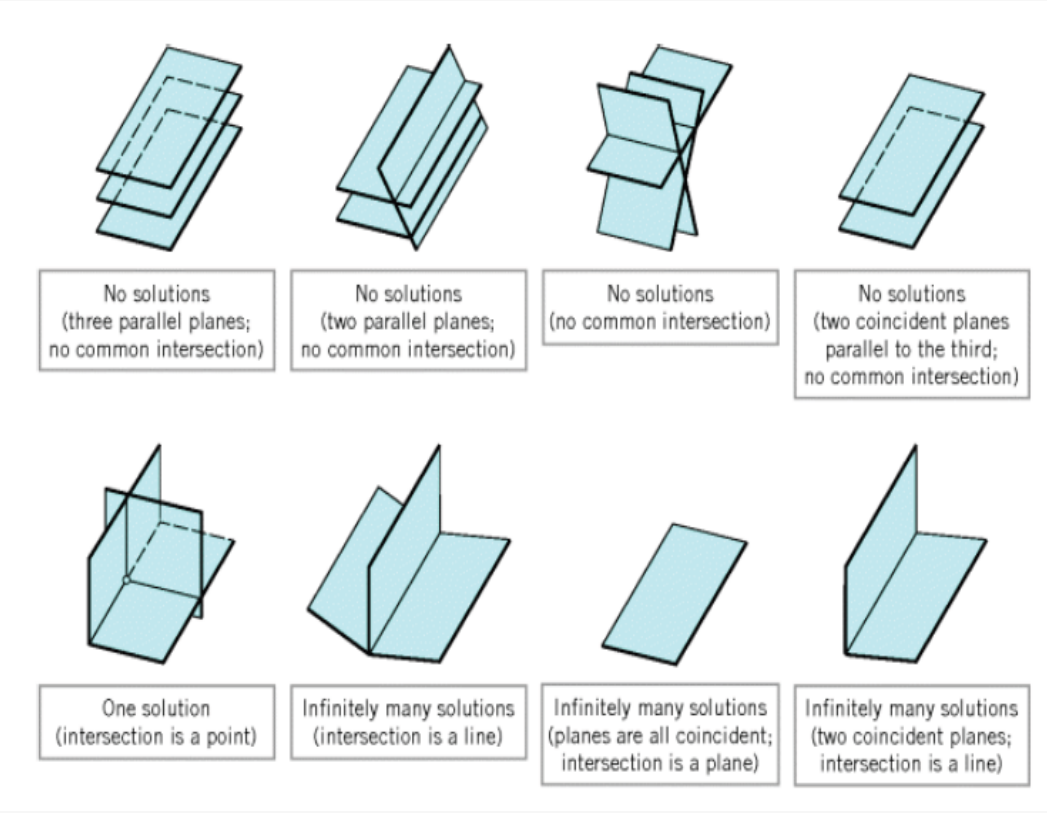
Similarly, for a linear system of three equations in three unknowns:

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned}$$

in which the graphs of the equations are planes,

the solutions of the system, if any, correspond to points where all three planes intersect.

So again we see that there are only three possibilities: no solutions, one solution, or infinitely many solutions



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Example: A Linear System with One Solution

Solve the linear system

$$x - y = 1$$
$$2x + y = 6$$

Solution We can eliminate x from the second equation by adding -2 times the first equation to the second. This yields the simplified system

$$x - y = 1$$
$$3y = 4$$

From the second equation we obtain $y = \frac{4}{3}$, and on substituting this value in the first equation we obtain $x = 1 + y = \frac{7}{3}$. Thus, the system has the unique solution

$$x = \frac{7}{3},\ y = \frac{4}{3}$$

Geometrically, this means that the lines represented by the equations in the system intersect at the single point $\left(\frac{7}{3}, \frac{4}{3}\right)$. We leave it for you to check this by graphing the lines.

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Example: A Linear System with No Solution

Solve the linear system

$$x + y = 4$$
$$3x + 3y = 6$$

Solution We can eliminate x from the second equation by adding -3 times the first equation to the second equation. This yields the simplified system

$$x + y = 4$$
$$0 = -6$$

The second equation is contradictory, so the given system has no solution. Geometrically, this means that the lines corresponding to the equations in the original system are parallel and distinct. We leave it for you to check this by graphing the lines or by showing that they have the same slope but different y-intercepts.

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Example 1: A Linear System with Infinitely Many Solutions

Solve the linear system

$$4x - 2y = 1$$
$$16x - 8y = 4$$

Solution We can eliminate x from the second equation by adding -4 times the first equation to the second. This yields the simplified system

$$4x - 2y = 1$$
$$0 = 0$$

The second equation does not impose any restrictions on x and y and hence can be omitted. Thus, the solutions of the system are those values of x and y that satisfy the single equation

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Example 1: A Linear System with Infinitely Many Solutions (cont.)

$$4x - 2y = 1 \tag{8}$$

Geometrically, this means the lines corresponding to the two equations in the original system coincide. One way to describe the solution set is to solve this equation for x in terms of y to obtain $x = \frac{1}{4} + \frac{1}{2}y$ and then assign an arbitrary value t (called a ***parameter***) to y. This allows us to express the solution by the pair of equations (called ***parametric equations***)

$$x = \frac{1}{4} + \frac{1}{2}t, \quad y = t$$

We can obtain specific numerical solutions from these equations by substituting numerical values for the parameter. For example, $t = 0$ yields the solution $\left(\frac{1}{4}, 0\right)$, $t = 1$ yields the solution $\left(\frac{3}{4}, 1\right)$, and $t = -1$ yields the solution $\left(-\frac{1}{4}, -1\right)$. You can confirm that these are solutions by substituting the coordinates into the given equations.

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Example 2: A Linear System with Infinitely Many Solutions

Solve the linear system

$$\begin{aligned} x - y + 2z &= 5 \\ 2x - 2y + 4z &= 10 \\ 3x - 3y + 6z &= 15 \end{aligned}$$

Solution This system can be solved by inspection, since the second and third equations are multiples of the first. Geometrically, this means that the three planes coincide and that those values of x, y, and z that satisfy the equation

$$x - y + 2z = 5 \tag{9}$$

automatically satisfy all three equations. Thus, it suffices to find the solutions of 9. We can do this by first solving 9 for x in terms of y and z, then assigning arbitrary values r and s (parameters) to these two variables, and then expressing the solution by the three parametric equations

$$x = 5 + r - 2s, \quad y = r, \quad z = s$$

Specific solutions can be obtained by choosing numerical values for the parameters r and s. For example, taking $r = 1$ and $s = 0$ yields the solution (6, 1,0).

Methods of solving systems of linear equations

Gaussian and Gauss-Jordan Elimination

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A system of linear equation can be represented as a matrix.
This is called the augmented matrix for the system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots \qquad\qquad & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

$$\Rightarrow \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{pmatrix}$$

Example : Find the augmented matrix of system of linear equation

$x_1 + x_2 + 2x_3 = 9$
$2x_1 - 3x_3 = 1$
$3x_1 + 6x_2 - 5x_3 = 0$

$$\Rightarrow \begin{pmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{pmatrix}$$

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Elementary row operations on a matrix

Three operations correspond to the following operations on the rows of the augmented matrix:

1. Multiply a row through by a nonzero constant. (cR_i)
2. Interchange two rows. ($R_i \leftrightarrow R_j$)
3. Add a constant times one row to another. ($R_i + cR_j$).

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Row echelon form.

A matrix that has the following properties is said to be in **reduced row echelon form:**

1. If a row does not consist entirely of zeros, then the first nonzero number in the row is a 1.
   We call this **a leading 1**.
2. If there are any rows that consist entirely of zeros, then they are grouped together at the bottom of the matrix.
3. In any two successive rows that do not consist entirely of zeros, the leading 1 in the lower row occurs farther to the right than the leading 1 in the higher row.
4. Each column that contains a leading 1 has zeros everywhere else in that column.

A matrix that has **the first three properties is said to be in row echelon form**

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Gauss-Jordan elimination.

A matrix that has the following properties is said to be in **reduced row echelon form**.

1. If a row does not consist entirely of zeros, then the first nonzero number in the row is a 1. We call this a leading 1.
2. If there are any rows that consist entirely of zeros, then they are grouped together at the bottom of the matrix.
3. In any two successive rows that do not consist entirely of zeros, the leading 1 in the lower row occurs farther to the right than the leading 1 in the higher row.
4. Each column that contains a leading 1 has zeros everywhere else in that column.

A matrix that has **the first three properties is said to be in row echelon form**

The procedure (or algorithm) to reduce a matrix into reduced row echelon form is called Gauss-Jordan elimination.

The procedure (or algorithm) to reduce a matrix into row echelon form followed by back substitutions is called Gaussian elimination.

Example 1 : Solve by Gauss elimination.

In the left column we solve a system of linear equations by operating on the equations in the system, and in the right column we solve the same system by operating on the rows of the augmented matrix.

$$\begin{aligned} x + y + 2z &= 9 \\ 2x + 4y - 3z &= 1 \\ 3x + 6y - 5z &= 0 \end{aligned}$$

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

Add −2 times the first equation to the second to obtain

$$\begin{aligned} x + y + 2z &= 9 \\ 2y - 7z &= -17 \\ 3x + 6y - 5z &= 0 \end{aligned}$$

Add −2 times the first row to the second to obtain

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 3 & 6 & -5 & 0 \end{bmatrix}$$

Add −3 times the first equation to the third to obtain

$$\begin{aligned} x + y + 2z &= 9 \\ 2y - 7z &= -17 \\ 3y - 11z &= -27 \end{aligned}$$

Add −3 times the first row to the third to obtain

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{bmatrix}$$

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Example 1 : Solve by Gauss elimination. (cont.)

Multiply the second equation by $\frac{1}{2}$ to obtain

$$\begin{aligned} x + y + 2z &= 9 \\ y - \tfrac{7}{2}z &= -\tfrac{17}{2} \\ 3y - 11z &= -27 \end{aligned}$$

Multiply the second row by $\frac{1}{2}$ to obtain

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{bmatrix}$$

Add -3 times the second equation to the third to obtain

$$\begin{aligned} x + y + 2z &= 9 \\ y - \tfrac{7}{2}z &= -\tfrac{17}{2} \\ -\tfrac{1}{2}z &= -\tfrac{3}{2} \end{aligned}$$

Add -3 times the second row to the third to obtain

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{bmatrix}$$

Multiply the third equation by -2 to obtain

$$\begin{aligned} x + y + 2z &= 9 \\ y - \tfrac{7}{2}z &= -\tfrac{17}{2} \\ z &= 3 \end{aligned}$$

Multiply the third row by -2 to obtain

$$\begin{bmatrix} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Example 1 : Solve by Gauss elimination (cont.)

Add −1 times the second equation to the first to obtain

$$\begin{aligned} x \quad &+ \tfrac{11}{2}z = \tfrac{35}{2} \\ y - \tfrac{7}{2}z &= -\tfrac{17}{2} \\ z &= 3 \end{aligned}$$

Add −1 times the second row to the first to obtain

$$\begin{bmatrix} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

We proceed with back substitutions.

From the equations $z = 3$ and $y - 7z/2 = -17/2$, we have $y = 2$.

From the equations $z = 3$ and $x + 11z/2 = 35/2$, we have $x = 1$.

Hence, the solution is $\boldsymbol{x = 1, y = 2, z = 3}$.

Example 2 : Solve by Gauss-Jordan elimination.

Solve the system of linear equation in the previous example by Gauss-Jordan elimination!

$$\begin{aligned} x + y + 2z &= 9 \\ 2x + 4y - 3z &= 1 \\ 3x + 6y - 5z &= 0 \end{aligned}$$

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Example 2 : Solve by Gauss-Jordan elimination (cont.)

Solution: Similar as in the previous example, we proceed until we have the augmented matrix in row echelon form.

$$\begin{aligned} x + \frac{11}{2}z &= \frac{35}{2} \\ y - \frac{7}{2}z &= -\frac{17}{2} \\ z &= 3 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Add $-\frac{11}{2}$ times the third equation to the first and $\frac{7}{2}$ times the third equation to the second to obtain

Add $-\frac{11}{2}$ times the third row to the first and $\frac{7}{2}$ times the third row to the second to obtain

$$\begin{aligned} x &= 1 \\ y &= 2 \\ z &= 3 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

The solution $x = 1,\ y = 2,\ z = 3$ is now evident.

Homogeneous System of Linear Equations

A system of linear equations is said to be ***homogeneous*** if the constant terms are all zero; that is, the system has the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= 0 \\ \vdots \qquad\quad & \vdots \qquad\quad \vdots \qquad\quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= 0 \end{aligned}$$

Every homogeneous system of linear equations is always consistent because all such systems at least have

$x_1=0$, $x_2=0$, ..., $x_n=0$ as a solution. This solution is called the ***trivial solution.***

If there are other solutions, they are called ***nontrivial solutions***.

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Homogeneous System of Linear Equations

Because a homogeneous linear system always has the trivial solution, there are only two possibilities for its solutions:

- The system has only the trivial solution.
- The system has infinitely many solutions in addition to the trivial solution.

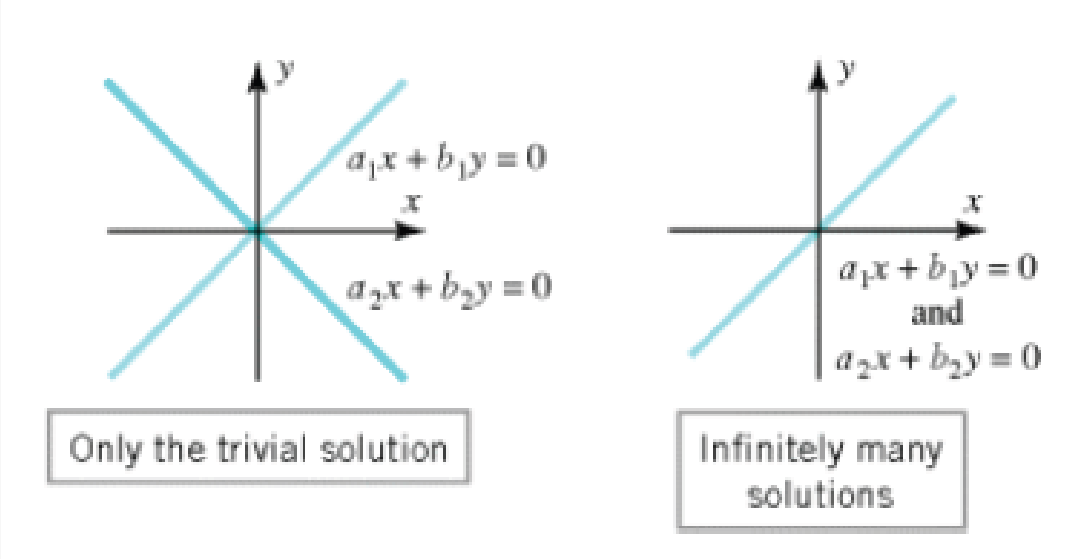
In the special case of a homogeneous linear system of two equations in two unknowns, say

$$a_1x + b_1y = 0 \quad (a_1, b_1 \text{ not both zero})$$
$$a_2x + b_2y = 0 \quad (a_2, b_2 \text{ not both zero})$$

the graphs of the equations are lines through the origin, and the trivial solution corresponds to the point of intersection at the origin.

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Illustration



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There is one case in which a homogeneous system is assured of having nontrivial solutions : whenever the system involves more unknowns than equations.

Example : Solving the homogeneous linear system

Use Gauss-Jordan elimination to solve the homogeneous linear system

$$\begin{aligned} x_1 + 3x_2 - 2x_3 \qquad + 2x_5 \qquad &= 0 \\ 2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 &= 0 \\ 5x_3 + 10x_4 \qquad + 15x_6 &= 0 \\ 2x_1 + 6x_2 \qquad + 8x_4 + 4x_5 + 18x_6 &= 0 \end{aligned} \tag{4}$$

Solution Observe first that the coefficients of the unknowns in this system are the same as those in Example 5; that is, the two systems differ only in the constants on the right side. The augmented matrix for the given homogeneous system is

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & 0 \\ 0 & 0 & 5 & 10 & 0 & 15 & 0 \\ 2 & 6 & 0 & 8 & 4 & 18 & 0 \end{bmatrix} \tag{5}$$

which is the same as the augmented matrix for the system in Example 5, except for zeros in the last column. Thus, the reduced row echelon form of this matrix will be the same as that of the augmented matrix in Example 5, except for the last column. However, a moment's reflection will make it evident that a column of zeros is not changed by an elementary row operation, so the reduced row echelon form of 5 is

$$\begin{bmatrix} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{6}$$

Example (Cont.)

The corresponding system of equations is

$$\begin{aligned} x_1 + 3x_2 \quad + 4x_4 + 2x_5 \quad &= 0 \\ x_3 + \ 2x_4 \qquad\quad &= 0 \\ x_6 &= 0 \end{aligned}$$

Solving for the leading variables we obtain

$$\begin{aligned} x_1 &= -3x_2 - 4x_4 - 2x_5 \\ x_3 &= -2x_4 \\ x_6 &= 0 \end{aligned} \tag{7}$$

If we now assign the free variables x_2, x_4, and x_5 arbitrary values r, s, and t, respectively, then we can express the solution set parametrically as

$$x_1 = -3r - 4s - 2t, \quad x_2 = r, \quad x_3 = -2s, \quad x_4 = s, \quad x_5 = t, \quad x_6 = 0$$

Note that the trivial solution results when $r = s = t = 0$.

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References

Howard Anton and Anton Kaul. (2019). Elementary Linear Algebra. 12. Wiley. New Jersey. ISBN: 978-1-119-40677-8

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Thank You

